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Imaging system, endoscope system, and control device

Abstract

An imaging system includes a camera device and a control device connected to each other by a power source line, a video signal line, and a ground line. An image sensor of the camera device receives, as a distal end voltage, a power source voltage output from the control device to the power source line. An output circuit of the camera device outputs a digital signal indicating a value of the distal end voltage to the video signal line. A calculation circuit of the control device calculates a resistance value of the power source line based on a value of the power source voltage, the value of the distal end voltage indicated by the digital signal, a current value of the power source line, a resistance value of the ground line, and a current value of the ground line.

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Patent No.	Application Date	Country	CPC
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Background/Summary

(1) Priority is claimed on U.S. Provisional Patent Application No. 63/406,528, filed on Sep. 14, 2022, the content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an imaging system, an endoscope system, and a control device.

Description of Related Art

(2) An endoscope system includes a camera device (scope) and a control device. The camera device and the control device are connected to each other by a power source line. An image sensor is mounted in the distal end of the camera device. The control device outputs a power source voltage used for driving the image sensor to the power source line. The power source voltage is transferred to the camera device by the power source line and is input to the camera device as a distal end voltage. Due to an influence of both a current flowing through the power source line and the resistance value of the power source line, a voltage drop is generated in the power source line. Therefore, the distal end voltage in the image sensor is lower than the power source voltage output from the control device.

(3) The amount of the current flowing through the power source line changes in accordance with the driving state of the image sensor. Therefore, the control device needs to output a power source

voltage having a high value to the power source line in light of the voltage drop generated in accordance with a change of the current. However, a power source voltage having a high value causes an increase of power consumption in the image sensor and an increase of the amount of heat generation in the power source line. Therefore, it is required that the value of the distal end voltage in the image sensor be monitored and a power source voltage having an optimal value be output to the power source line.

(4) A method of calculating the resistance value of the power source line and calculating the value of the distal end voltage in accordance with the following Expression (1) has been considered in order to monitor the value of the distal end voltage.

$$V_{cis} = V_{out} - R_c \cdot I_c \quad (1)$$

(5) A voltage value V_{cis} in Expression (1) indicates the value of the distal end voltage, and a voltage value V_{out} in Expression (1) indicates the value of the power source voltage. A resistance value R_c in Expression (1) indicates the resistance value of the power source line, and a current value I_c in Expression (1) indicates the value of the current flowing through the power source line.

(6) The resistance value of the power source line changes in accordance with an individual difference of the power source line, passage of time, a change of the temperature, and the like. In the above-described method, it is necessary to calculate the resistance value of the power source line in order to properly control the power source voltage.

(7) A technique disclosed in PCT International Publication No. WO2022/013953 provides a method of calculating a resistance value of a power source line. In the technique, a camera device generates a digital value of a distal end voltage by performing analog-to-digital (AD) conversion. The camera device transmits the generated digital value to a control device. The control device receives the digital value and calculates a resistance value of a power source line in accordance with the following Expression (2).

$$R_c = (V_{out} - V_{cis}) / I_c \quad (2)$$

(8) A resistance value R_c in Expression (2) indicates the resistance value of the power source line, and a current value I_c in Expression (2) indicates the value of the current flowing through the power source line. A voltage value V_{out} in Expression (2) indicates the value of the power source voltage, and a voltage value V_{cis} in Expression (2) indicates the value of the distal end voltage.

SUMMARY OF THE INVENTION

(9) According to a first aspect of the present invention, an imaging system includes a camera device and a control device connected to each other by a power source line, a video signal line, and a ground line. The camera device includes an image sensor, a voltage measurement circuit, and an output circuit. The image sensor receives, as a distal end voltage, a power source voltage output from the control device to the power source line and generates a video signal by using the distal end voltage. The voltage measurement circuit measures a value of the distal end voltage. The output circuit outputs the video signal to the video signal line and outputs a digital signal indicating the value of the distal end voltage to the video signal line. The control device includes a voltage generation circuit, a current measurement unit circuit, a reception circuit, a calculation circuit, and a voltage adjustment circuit. The voltage generation circuit generates the power source voltage and outputs the generated power source voltage to the power source line. The current measurement circuit measures a current value of the power source line. The reception circuit receives the video signal and the digital signal output from the camera device to the video signal line. The calculation circuit calculates a resistance value of the power source line based on a value of the power source voltage, the value of the distal end voltage indicated by the digital signal, the current value of the power source line, a resistance value of the ground line, and a current value of the ground line. The voltage adjustment circuit adjusts a value of the power source voltage based on the resistance value of the power source line, the current value of the power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the distal end voltage nears a target value.

(10) According to a second aspect of the present invention, in the first aspect, the power source line may include a first power source line and a second power source line. The image sensor may receive, as a first distal end voltage, a first power source voltage output from the control device to the first power source line and may receive, as a second distal end voltage, a second power source voltage output from the control device to the second power source line. A value of the second power source voltage may be different from a value of the first power source voltage. The voltage measurement circuit may measure a value of the first distal end voltage and a value of the second distal end voltage. The output circuit may output a first digital signal indicating the value of the first distal end voltage and a second digital signal indicating the value of the second distal end voltage to the video signal line. The voltage generation circuit may generate the first power source voltage and the second power source voltage. The voltage generation circuit may output the generated first power source voltage to the first power source line and may output the generated second power source voltage to the second power source line. The current measurement circuit may measure a current value of the first power source line and a current value of the second power source line. The reception circuit may receive the first digital signal and the second digital signal. The calculation circuit may calculate a resistance value of the first power source line based on the value of the first power source voltage, the value of the first distal end voltage indicated by the first digital signal, the current value of the first power source line, the resistance value of the ground line, and the current value of the ground line. The calculation circuit may calculate a resistance value of the second power source line based on the value of the second power source voltage, the value of the second distal end voltage indicated by the second digital signal, the current value of the second power source line, the resistance value of the ground line, and the current value of the ground line. The voltage adjustment circuit may adjust the value of the first power source voltage based on the resistance value of the first power source line, the current value of the first power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the first distal end voltage nears a first target value. The voltage adjustment circuit may adjust the value of the second power source voltage based on the resistance value of the second power source line, the current value of the second power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the second distal end voltage nears a second target value.

(11) According to a third aspect of the present invention, in the second aspect, the calculation circuit may calculate the resistance value of the ground line based on the value of the first power source voltage, the value of the first distal end voltage indicated by the first digital signal, and the current value of the ground line.

(12) According to a fourth aspect of the present invention, in the second aspect, the voltage generation circuit may output the first power source voltage to the first power source line in a first period and a second period and may output the second power source voltage to the second power source line in the first period and the second period. A current value of the first power source line in the second period may be the same as a current value of the first power source line in the first period. A current value of the second power source line in the second period may be different from a current value of the second power source line in the first period. The calculation circuit may calculate the resistance value of the ground line based on the value of the first power source voltage in each of the first period and the second period, the value of the first distal end voltage indicated by the first digital signal in each of the first period and the second period, and the current value of the ground line in each of the first period and the second period.

(13) According to a fifth aspect of the present invention, in the fourth aspect, the first period and the second may be included in a blanking period of the image sensor.

(14) According to a sixth aspect of the present invention, in the second aspect, the calculation circuit may calculate the current value of the ground line based on the current value of the first power source line and the current value of the second power source line.

- (15) According to a seventh aspect of the present invention, in the second aspect, one of the first distal end voltage and the second distal end voltage is provided to a digital circuit included in the image sensor. The other of the first distal end voltage and the second distal end voltage may be provided to an analog circuit included in the image sensor.
- (16) According to an eighth aspect of the present invention, in the first aspect, the resistance value of the ground line may be the same as the resistance value of the power source line.
- (17) According to a ninth aspect of the present invention, an endoscope system includes a scope to be inserted into a living body and includes the imaging device. The camera device is disposed in a distal end of the scope.
- (18) According to a tenth aspect of the present invention, a control device is used in an imaging system including a camera device and the control device connected to each other by a power source line, a video signal line, and a ground line. The control device includes a voltage generation circuit, a current measurement circuit, a reception circuit, a calculation circuit, and a voltage adjustment circuit. The voltage generation circuit generates a power source voltage and outputs the generated power source voltage to the power source line. The current measurement circuit measures a current value of the power source line. The reception circuit receives a video signal and a digital signal output from the camera device to the video signal line. The video signal is generated by an image sensor of the camera device. The image sensor receives, as a distal end voltage, the power source voltage output from the control device to the power source line and generates the video signal by using the distal end voltage. The digital signal indicates a value of the distal end voltage. The calculation circuit calculates a resistance value of the power source line based on a value of the power source voltage, the value of the distal end voltage indicated by the digital signal, the current value of the power source line, a resistance value of the ground line, and a current value of the ground line. The voltage adjustment circuit adjusts a value of the power source voltage based on the resistance value of the power source line, the current value of the power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the distal end voltage nears a target value.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram showing a configuration of an endoscope system according to a first embodiment of the present invention.
- (2) FIG. 2 is a block diagram showing a configuration of the endoscope system according to the first embodiment of the present invention.
- (3) FIG. 3 is a flow chart showing a procedure of an operation of a control unit included in the endoscope system according to the first embodiment of the present invention.
- (4) FIG. 4 is a timing chart showing waveforms of currents flowing through a transmission cable included in the endoscope system according to the first embodiment of the present invention.
- (5) FIG. 5 is a flow chart showing a procedure of an operation of a camera device included in the endoscope system according to the first embodiment of the present invention.
- (6) FIG. 6 is a flow chart showing a procedure of an operation of a control unit included in an endoscope system according to a modified example of the first embodiment of the present invention.
- (7) FIG. 7 is a timing chart showing waveforms of currents flowing through a transmission cable included in the endoscope system according to the modified example of the first embodiment of the present invention.
- (8) FIG. 8 is a flow chart showing a procedure of an operation of a camera device included in the endoscope system according to the modified example of the first embodiment of the present invention.

invention.

(9) FIG. 9 is a flow chart showing a procedure of an operation of a control unit included in an endoscope system according to a second embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(10) Hereinafter, embodiments of the present invention will be described with reference to the drawings. Each of the embodiments will be described in detail by using an endoscope system as an example of an imaging system.

First Embodiment

(11) FIG. 1 shows a configuration of an endoscope system 1 according to a first embodiment of the present invention. The endoscope system 1 shown in FIG. 1 includes an endoscope insertion unit 2, a transmission cable 3, an operation unit 4, a connector unit 5, a control unit 6, and a display device 7. The endoscope insertion unit 2, the transmission cable 3, the operation unit 4, and the connector unit 5 constitute a scope 8.

(12) The endoscope insertion unit 2 includes an insertion unit 2a. The insertion unit 2a is part of the transmission cable 3. The insertion unit 2a is to be inserted inside a living body, which is a subject. The endoscope insertion unit 2 generates a video signal by imaging the inside of the subject. The endoscope insertion unit 2 outputs the generated video signal to the control unit 6. A camera device 9 shown in FIG. 2 is disposed in a distal end 2b of the insertion unit 2a. In the insertion unit 2a, the operation unit 4 is connected to the end part opposite the distal end 2b. The operation unit 4 receives various operations for the endoscope insertion unit 2 from a user.

(13) The transmission cable 3 connects the camera device 9 and the connector unit 5. The video signal generated by the camera device 9 is output to the connector unit 5 via the transmission cable 3.

(14) The connector unit 5 is connected to the endoscope insertion unit 2 and the control unit 6. The connector unit 5 performs predetermined processing on the video signal output from the endoscope insertion unit 2. The connector unit 5 outputs the video signal to the control unit 6.

(15) The control unit 6 performs image processing on the video signal output from the connector unit 5. Furthermore, the control unit 6 centrally controls the entire endoscope system 1.

(16) The display device 7 displays a video based on the video signal processed by the control unit 6. In addition, the display device 7 displays various kinds of information related to the endoscope system 1.

(17) FIG. 2 shows an internal configuration of the endoscope system 1. The endoscope system 1 shown in FIG. 2 includes the camera device 9, the connector unit 5, and the control unit 6. The camera device 9 is disposed in the distal end 2b of the scope 8. The connector unit 5 and the control unit 6 constitute a control device 10. The operation unit 4 is not shown in FIG. 2. The transmission cable 3 shown in FIG. 1 includes a video signal line 30, a power source line 31, a power source line 32, and a ground line 33 shown in FIG. 2.

(18) The endoscope system 1 includes a light source device that generates illumination light emitted to the subject. The light source device is omitted in FIG. 2. The camera device 9 includes an image sensor 90. The image sensor 90 includes a pixel unit 91, an AD conversion unit 92, and a signal-processing unit 93 (output circuit).

(19) The connector unit 5 includes an analog front end (AFE) unit 50 (reception circuit) and a signal-processing unit 51. All or part of the configuration of the connector unit 5 shown in FIG. 2 may be disposed in the operation unit 4 or the control unit 6.

(20) The control unit 6 includes a power source unit 60 (voltage generation circuit), an image-processing unit 61, a voltage measurement unit 62 (voltage measurement circuit), a current measurement unit 63 (current measurement circuit), a resistance calculation unit 64 (calculation circuit), and a power source control unit 65 (power source adjustment circuit). All or part of the configuration of the control unit 6 shown in FIG. 2 may be disposed in the operation unit 4 or the connector unit 5.

(21) A schematic configuration of the endoscope system **1** will be described. The camera device **9** and the control unit **6** are connected to each other by the power source line **31**, the power source line **32**, the video signal line **30**, and the ground line **33**. The image sensor **90** receives, as a distal end voltage V_{cis_dig} (V_{cis_ana}), a power source voltage V_{out_dig} (V_{out_ana}) output from the control unit **6** to the power source line **31** (power source line **32**) and generates a video signal by using the distal end voltage V_{cis_dig} (V_{cis_ana}). The AD conversion unit **92** measures a value of the distal end voltage V_{cis_dig} (V_{cis_ana}). The signal-processing unit **93** outputs the video signal to the video signal line **30** and outputs a digital signal indicating the value of the distal end voltage V_{cis_dig} (V_{cis_ana}) to the video signal line **30** (power source line **32**). The power source unit **60** generates the power source voltage V_{out_dig} (V_{out_ana}) and outputs the generated power source voltage V_{out_dig} (V_{out_ana}) to the power source line **31**. The current measurement unit **63** measures a current value of the power source line **31** (power source line **32**). The AFE unit **50** receives the video signal and the digital signal output from the camera device **9** to the video signal line **30**. The resistance calculation unit **64** calculates a resistance value of the power source line **31** (power source line **32**) based on the value of the power source voltage V_{out_dig} (V_{out_ana}), the value of the distal end voltage V_{cis_dig} (V_{cis_ana}) indicated by the digital signal, the current value of the power source line **31** (power source line **32**), a resistance value of the ground line **33**, and a current value of the ground line **33**. The power source control unit **65** adjusts the value of the power source voltage V_{out_dig} (V_{out_ana}) based on the resistance value of the power source line **31** (power source line **32**), the current value of the power source line **31** (power source line **32**), the resistance value of the ground line **33**, and the current value of the ground line **33** such that the value of the distal end voltage V_{cis_dig} (V_{cis_ana}) nears a target value.

(22) A detailed configuration of the endoscope system **1** will be described. For example, the power source unit **60** includes a voltage regulator. The power source unit **60** generates the power source voltage V_{out_dig} and the power source voltage V_{out_ana} . The power source voltage V_{out_dig} is a direct-current (DC) voltage to be provided to a digital circuit included in the image sensor **90**. The power source voltage V_{out_ana} is a DC voltage to be provided to an analog circuit included in the image sensor **90**. In general, the value of the power source voltage V_{out_dig} is greater than that of the power source voltage V_{out_ana} . The value of the power source voltage V_{out_ana} is greater than that (0 V) of a ground voltage (reference voltage). For example, the value of the power source voltage V_{out_dig} is 1 V or the like, and the value of the power source voltage V_{out_ana} is 2.7 V or the like.

(23) The power source unit **60** is connected to the power source line **31** and the power source line **32**. The power source unit **60** outputs the power source voltage V_{out_dig} to the power source line **31** and outputs the power source voltage V_{out_ana} to the power source line **32**. The power source line **31** and the power source line **32** are signal lines disposed in the transmission cable **3**. The power source line **31** transfers the power source voltage V_{out_dig} to the camera device **9**. The power source line **32** transfers the power source voltage V_{out_ana} to the camera device **9**.

(24) The power source voltage V_{out_dig} transferred by the power source line **31** and the power source voltage V_{out_ana} transferred by the power source line **32** are input to the camera device **9**. The image sensor **90** is connected to the power source line **31** and the power source line **32**. The power source voltage V_{out_dig} is provided as the distal end voltage V_{cis_dig} to each circuit in the image sensor **90**. For example, the distal end voltage V_{cis_dig} is output to the signal-processing unit **93**. In addition, the distal end voltage V_{cis_dig} is output to a register that stores a value used for controlling the image sensor **90**. The power source voltage V_{out_ana} is provided as the distal end voltage V_{cis_ana} to each circuit in the image sensor **90**. For example, the distal end voltage V_{cis_ana} is output to the pixel unit **91** and the AD conversion unit **92**.

(25) A current I_{dig} flows through the power source line **31**, and a current I_{ana} flows through the power source line **32**. A voltage drop is generated due to the DC resistance of the power source line **31**, and the distal end voltage V_{cis_dig} is attenuated. Therefore, the value of the distal end voltage

Vcis_dig is less than that of the power source voltage Vout_dig in the control unit **6**. Similarly, a voltage drop is generated due to the DC resistance of the power source line **32**, and the distal end voltage Vcis_ana is attenuated. Therefore, the value of the distal end voltage Vcis_ana is less than that of the power source voltage Vout_ana in the control unit **6**.

(26) The ground line **33** is a signal line disposed in the transmission cable **3**. The ground line **33** transfers the ground voltage from the control unit **6** to the camera device **9**. The image sensor **90** is connected to the ground line **33**. The ground voltage transferred by the ground line **33** is input to the camera device **9** and is provided to each circuit in the image sensor **90**. A current Ig flows through the ground line **33**. A voltage drop is generated due to the DC resistance of the ground line **33**. Therefore, the value of the ground voltage in the camera device **9** is different from that in the control unit **6**.

(27) The image sensor **90** is a complementary metal-oxide semiconductor (CMOS) sensor or the like. The pixel unit **91** includes two or more pixels and is driven based on the distal end voltage Vcis_ana. The pixel unit **91** generates a pixel signal having a voltage generated based on the distal end voltage Vcis_ana and the ground voltage. The pixel unit **91** outputs the pixel signal in a video output period and stops the output of the pixel signal in a blanking period.

(28) The AD conversion unit **92** performs AD conversion on the pixel signal output from the pixel unit **91** and generates a digital pixel signal. The AD conversion unit **92** performs the AD conversion on the distal end voltage Vcis_dig and generates a digital signal indicating a value of the distal end voltage Vcis_dig. The AD conversion unit **92** performs the AD conversion on the distal end voltage Vcis_ana and generates a digital signal indicating a value of the distal end voltage Vcis_ana.

(29) After the power source of the endoscope system **1** is turned on and the power source unit **60** starts generation of the power source voltage Vout_dig and the power source voltage Vout_ana, a resistance calculation period for calculating the resistance value (ground resistance value) of the ground line **33** is set. For example, the AD conversion unit **92** generates the digital signal in the resistance calculation period before the pixel unit **91** generates the pixel signal. The AD conversion unit **92** may generate the digital signal in the blanking period.

(30) The signal-processing unit **93** processes the digital pixel signal output from the AD conversion unit **92** and generates a video signal. For example, the signal-processing unit **93** performs noise reduction, amplification processing, and the like on the digital pixel signal. The signal-processing unit **93** outputs the video signal to the video signal line **30**. In addition, the signal-processing unit **93** outputs a digital signal indicating a value of each of the power source voltage Vout_dig and the power source voltage Vout_ana to the video signal line **30**.

(31) At least one of the AD conversion unit **92** and the signal-processing unit **93** may be disposed in a different substrate from that in which the image sensor **90** is disposed. In other words, at least one of the AD conversion unit **92** and the signal-processing unit **93** may be disposed outside the image sensor **90**.

(32) The video signal line **30** is a signal line disposed in the transmission cable **3**. The video signal line **30** transfers the video signal and the digital signal output from the signal-processing unit **93** to the connector unit **5**.

(33) The AFE unit **50** receives the video signal and the digital signal transferred by the video signal line **30**. The AFE unit **50** performs processing such as noise elimination on the video signal and the digital signal. The signal-processing unit **51** performs gain-up processing or the like on the video signal and the digital signal output from the AFE unit **50**. The signal-processing unit **51** outputs the video signal to the image-processing unit **61** and outputs the digital signal to the resistance calculation unit **64**.

(34) The image-processing unit **61** performs demosaicing processing, white balance adjustment processing, γ correction processing, and the like on the video signal. The image-processing unit **61** output the video signal to the display device **7**.

(35) The voltage measurement unit **62** measures a value of the power source voltage Vout_dig

output from the power source unit **60** to the power source line **31** and measures a value of the power source voltage Vout_ana output from the power source unit **60** to the power source line **32**. The voltage measurement unit **62** outputs the value of the power source voltage Vout_dig and the value of the power source voltage Vout_ana to the resistance calculation unit **64**.

(36) The current measurement unit **63** measures a value of the current Idig flowing through the power source line **31** and measures a value of the current Iana flowing through the power source line **32**. The current measurement unit **63** outputs a current value of the power source line **31** and a current value of the power source line **32** to the resistance calculation unit **64**.

(37) The resistance calculation unit **64** calculates the resistance value of the ground line **33** based on the value of the power source voltage Vout_dig, the value of the distal end voltage Vcis_dig indicated by the digital signal, and the current value of the ground line **33**. The resistance calculation unit **64** can calculate the current value of the ground line **33** by adding the current value of the power source line **31** and the current value of the power source line **32**. In a case in which the ground line **33** is constituted by a single signal line and a current as the ground current flows only through the ground line **33**, the camera device **9** may include a current measurement circuit that measures the ground current.

(38) The resistance calculation unit **64** calculates the resistance value of the power source line **31** based on the value of the power source voltage Vout_dig, the value of the distal end voltage Vcis_dig indicated by the digital signal, the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33**. In addition, the resistance calculation unit **64** calculates the resistance value of the power source line **32** based on the value of the power source voltage Vout_ana, the value of the distal end voltage Vcis_ana indicated by the digital signal, the current value of the power source line **32**, the resistance value of the ground line **33**, and the current value of the ground line **33**. A method of calculating the resistance value of the ground line **33**, the resistance value of the power source line **31**, and the resistance value of the power source line **32** will be described later. The resistance calculation unit **64** outputs the resistance value of the power source line **31** and the resistance value of the power source line **32** to the power source control unit **65**.

(39) The resistance calculation unit **64** is a digital circuit including at least one of a processor and a logic circuit. For example, the processor is a central processing unit (CPU). For example, the logic circuit is at least one of an application-specific integrated circuit (ASIC) or a field-programmable gate array (FPGA). The resistance calculation unit **64** may include one or a plurality of processors. The resistance calculation unit **64** may include one or a plurality of logic circuits.

(40) The resistance calculation unit **64** may read a program and execute the read program. The program includes commands defining the operations of the resistance calculation unit **64**. In other words, the functions of the resistance calculation unit **64** may be realized by software. The program may be transmitted from a computer storing the program to the endoscope system **1** through a transmission medium or transmission waves in a transmission medium. The “transmission medium” transmitting the program is a medium having a function of transmitting information. The medium having the function of transmitting information includes a network (communication network) such as the Internet and a communication circuit line (communication line) such as a telephone line. The program described above may realize some of the functions described above. In addition, the program described above may be a differential file (differential program). The functions described above may be realized by a combination of a program that has already been recorded in a computer and a differential program.

(41) The power source control unit **65** controls the power source unit **60** based on the resistance value of the power source line **31**, the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33**. By doing this, the power source control unit **65** adjusts the value of the power source voltage Vout_dig to be generated by the power source unit **60**. In addition, the power source control unit **65** controls the power source

unit **60** based on the resistance value of the power source line **32**, the current value of the power source line **32**, the resistance value of the ground line **33**, and the current value of the ground line **33**. By doing this, the power source control unit **65** adjusts the value of the power source voltage V_{out_ana} to be generated by the power source unit **60**.

(42) The transmission cable **3** may include three or more power source lines. Two of the three or more power source lines correspond to the power source line **31** and the power source line **32**.

(43) FIG. **3** shows a procedure of an operation of the control unit **6** related to adjustment of the power source voltage V_{out_dig} and the power source voltage V_{out_ana} . The operation of the control unit **6** will be described by using FIG. **3**.

(44) The resistance calculation period includes a first period and a second period. The power source unit **60** generates the power source voltage V_{out_dig} and the power source voltage V_{out_ana} in the first period. The power source unit **60** outputs the power source voltage V_{out_dig} to the power source line **31** and outputs the power source voltage V_{out_ana} to the power source line **32** (Step **S100**).

(45) After Step **S100**, the voltage measurement unit **62** measures a value V_{out_dig1} of the power source voltage V_{out_dig} in the power source line **31** and measures a value V_{out_ana1} of the power source voltage V_{out_ana} in the power source line **32**. The voltage measurement unit **62** outputs the value V_{out_dig1} and the value V_{out_ana1} to the resistance calculation unit **64** (Step **S105**).

(46) After Step **S105**, the current measurement unit **63** measures a current value I_{dig1} of the power source line **31** and a current value I_{ana1} of the power source line **32** and outputs the current value I_{dig1} and the current value I_{ana1} to the resistance calculation unit **64** (Step **S110**).

(47) After Step **S110**, the AFE unit **50** receives a digital signal indicating a value V_{cis_dig1} of the distal end voltage V_{cis_dig} in the first period and processes the digital signal. In addition, the AFE unit **50** receives a digital signal indicating a value V_{cis_ana1} of the distal end voltage V_{cis_ana} in the first period and processes the digital signal. Each digital signal processed by the AFE unit **50** is output to the signal-processing unit **51**. The signal-processing unit **51** processes each digital signal and outputs each digital signal to the resistance calculation unit **64** (Step **S115**).

(48) After Step **S115**, the resistance calculation unit **64** calculates a current value I_g1 of the ground line **33** by adding the current value I_{dig1} of the power source line **31** and the current value I_{ana1} of the power source line **32** (Step **S120**).

(49) The first period is completed, and the second period is started. The power source unit **60** generates the power source voltage V_{out_dig} and the power source voltage V_{out_ana} in the second period. The power source unit **60** outputs the power source voltage V_{out_dig} to the power source line **31** and outputs the power source voltage V_{out_ana} to the power source line **32** (Step **S125**).

(50) After Step **S125**, the voltage measurement unit **62** measures a value V_{out_dig2} of the power source voltage V_{out_dig} in the power source line **31** and outputs the value V_{out_dig2} to the resistance calculation unit **64** (Step **S130**).

(51) After Step **S130**, the current measurement unit **63** measures a current value I_{dig2} of the power source line **31** and a current value I_{ana2} of the power source line **32** and outputs the current value I_{dig2} and the current value I_{ana2} to the resistance calculation unit **64** (Step **S135**).

(52) After Step **S135**, the AFE unit **50** receives a digital signal transferred by the video signal line **30** and processes the digital signal. The digital signal indicates a value V_{cis_dig2} of the distal end voltage V_{cis_dig} in the second period. The digital signal is output to the signal-processing unit **51**. The signal-processing unit **51** processes the digital signal and outputs the digital signal to the resistance calculation unit **64** (Step **S140**).

(53) After Step **S140**, the resistance calculation unit **64** calculates a current value I_g2 of the ground line **33** by adding the current value I_{dig2} of the power source line **31** and the current value I_{ana2} of the power source line **32** (Step **S145**).

(54) FIG. **4** shows waveforms of the current I_{dig} , the current I_{ana} , and the current I_g . The horizontal axis in FIG. **4** indicates time, and the vertical axis in FIG. **4** indicates a current value.

(55) The power source unit **60** outputs the power source voltage V_{out_dig} to the power source line **31** such that a value I_{dig1} of the current I_{dig} in a first period **T1** and a value I_{dig2} of the current I_{dig} in a second period **T2** are the same. For example, a value V_{out_dig2} of the power source voltage V_{out_dig} in the second period **T2** is the same as a value V_{out_dig1} of the power source voltage V_{out_dig} in the first period **T1**.

(56) The power source unit **60** outputs the power source voltage V_{out_ana} to the power source line **32** such that a value I_{ana1} of the current I_{ana} in the first period **T1** and a value I_{ana2} of the current I_{ana} in the second period **T2** are different. For example, a value V_{out_ana2} of the power source voltage V_{out_ana} in the second period **T2** is different from a value V_{out_ana1} of the power source voltage V_{out_ana} in the first period **T1**.

(57) In the example shown in FIG. 4, the value I_{ana2} of the current I_{ana} in the second period **T2** is greater than the value I_{ana1} of the current I_{ana} in the first period **T1**. The value I_{ana2} of the current I_{ana} in the second period **T2** may be less than the value I_{ana1} of the current I_{ana} in the first period **T1**.

(58) The current I_g is the sum of the current I_{dig} and the current I_{ana} . The value of the current I_g in the first period **T1** is I_{g1} , and the value of the current I_g in the second period **T2** is I_{g2} .

(59) After Step **S145**, the resistance calculation unit **64** calculates a resistance value R_g of the ground line **33** based on the values V_{out_dig1} and V_{out_dig2} of the power source voltage V_{out_dig} , the values V_{cis_dig1} and V_{cis_dig2} of the distal end voltage V_{cis_dig} , and the current values I_{g1} and I_{g2} of the ground line **33** (Step **S150**).

(60) The following Expression (4) shows the value V_{out_dig1} of the power source voltage V_{out_dig} in the first period.

$$V_{sub.out_dig1} = R_{sub.dig} I_{sub.dig1} + V_{sub.cis_dig1} + R_{sub.g} I_{sub.g1} \quad (4)$$

(61) A resistance value R_{dig} in Expression (4) indicates the resistance value of the power source line **31**, and a current value I_{dig1} in Expression (4) indicates the current value of the power source line **31** in the first period. A voltage value V_{cis_dig1} in Expression (4) indicates the value of the distal end voltage V_{cis_dig} in the first period. A resistance value R_g in Expression (4) indicates the resistance value of the ground line **33**, and a current value I_{g1} in Expression (4) indicates the current value of the ground line **33** in the first period.

(62) The following Expression (5) shows the value V_{out_dig2} of the power source voltage V_{out_dig} in the second period.

$$V_{sub.out_dig2} = R_{sub.dig} I_{sub.dig2} + V_{sub.cis_dig2} + R_{sub.g} I_{sub.g2} \quad (5)$$

(63) A resistance value R_{dig} in Expression (5) indicates the resistance value of the power source line **31**, and a current value I_{dig2} in Expression (5) indicates the current value of the power source line **31** in the second period. A voltage value V_{cis_dig2} in Expression (5) indicates the value of the distal end voltage V_{cis_dig} in the second period. A resistance value R_g in Expression (5) indicates the resistance value of the ground line **33**, and a current value I_{g2} in Expression (5) indicates the current value of the ground line **33** in the second period.

(64) The current value I_{dig2} is the same as the current value I_{dig1} . Accordingly, the following Expression (6) showing a difference between Expression (4) and Expression (5) is obtained.

$$V_{sub.out_dig1} - V_{sub.out_dig2} = (V_{sub.cis_dig1} - V_{sub.cis_dig2}) + R_{sub.g} (I_{sub.g1} - I_{sub.g2}) \quad (6)$$

(65) By modifying Expression (6), the following Expression (7) is obtained. The resistance calculation unit **64** calculates the resistance value R_g of the ground line **33** in accordance with Expression (7) in Step **S150**.

$$R_g = \frac{(V_{out_dig1} - V_{out_dig2}) - (V_{cis_dig1} - V_{cis_dig2})}{I_{g1} - I_{g2}} \quad (7)$$

(67) After Step **S150**, the resistance calculation unit **64** calculates the resistance value R_{dig} of the power source line **31** based on the value of the power source voltage V_{out_dig} , the value of the distal end voltage V_{cis_dig} , the current value of the power source line **31**, the resistance value R_g

of the ground line 33, and the current value I_g of the ground line 33. In addition, the resistance calculation unit 64 calculates the resistance value R_{ana} of the power source line 32 based on the value of the power source voltage V_{out_ana} , the value of the distal end voltage V_{cis_ana} , the current value of the power source line 32, the resistance value R_g of the ground line 33, and the current value I_g of the ground line 33. The resistance calculation unit 64 outputs the resistance value R_{dig} of the power source line 31 and the resistance value R_{ana} of the power source line 32 to the power source control unit 65 (Step S155).

(68) By modifying Expression (4), the following Expression (8) is obtained. The resistance calculation unit 64 calculates the resistance value R_{dig} of the power source line 31 in accordance with Expression (8) in Step S155.

$$(69) \quad R_{dig} = \frac{V_{out_dig1} - V_{cis_dig1} - R_g I_{g1}}{I_{dig1}} \quad (8)$$

(70) Expression (8) uses the value V_{out_dig1} of the power source voltage V_{out_dig} in the first period, the value V_{cis_dig1} of the distal end voltage V_{cis_dig} in the first period, the resistance value R_g of the ground line 33, the current value I_{g1} of the ground line 33 in the first period, and the current value I_{dig1} of the power source line 31 in the first period. The resistance calculation unit 64 may calculate the resistance value R_{dig} of the power source line 31 by using the value V_{out_dig2} of the power source voltage V_{out_dig} in the second period, the value V_{cis_dig2} of the distal end voltage V_{cis_dig} in the second period, the resistance value R_g of the ground line 33, the current value I_{g2} of the ground line 33 in the second period, and the current value I_{dig2} of the power source line 31 in the second period.

(71) The following Expression (9) is a similar expression to Expression (8). The resistance calculation unit 64 calculates the resistance value R_{ana} of the power source line 32 in accordance with Expression (9) in Step S155.

$$(72) \quad R_{ana} = \frac{V_{out_ana1} - V_{cis_ana1} - R_g I_{g1}}{I_{ana1}} \quad (9)$$

(73) Expression (9) uses the value V_{out_ana1} of the power source voltage V_{out_ana} in the first period, the value V_{cis_ana1} of the distal end voltage V_{cis_ana} in the first period, the resistance value R_g of the ground line 33, the current value I_{g1} of the ground line 33 in the first period, and the current value I_{ana1} of the power source line 32 in the first period. The resistance calculation unit 64 may calculate the resistance value R_{ana} of the power source line 32 by using the value V_{out_ana2} of the power source voltage V_{out_ana} in the second period, the value V_{cis_ana2} of the distal end voltage V_{cis_ana} in the second period, the resistance value R_g of the ground line 33, the current value I_{g2} of the ground line 33 in the second period, and the current value I_{ana2} of the power source line 32 in the second period.

(74) After Step S155, the power source control unit 65 controls the power source unit 60 based on the resistance value R_{dig} of the power source line 31, the current value of the power source line 31, the resistance value R_g of the ground line 33, and the current value of the ground line 33. By doing this, the power source control unit 65 adjusts the value of the power source voltage V_{out_dig} . In addition, the power source control unit 65 controls the power source unit 60 based on the resistance value R_{ana} of the power source line 32, the current value of the power source line 32, the resistance value R_g of the ground line 33, and the current value of the ground line 33. By doing this, the power source control unit 65 adjusts the value of the power source voltage V_{out_ana} (Step S160).

(75) The following Expression (10) indicates a value V_{out_digc} of the power source voltage V_{out_dig} . The power source control unit 65 adjusts the value V_{out_digc} of the power source voltage V_{out_dig} in accordance with Expression (10) in Step S160.

$$V_{sub.out_digc} = R_{sub.dig} I_{sub.dig} + V_{sub.tar_dig} + R_{sub.g} I_{sub.g} \quad (1)$$

(76) A resistance value R_{dig} in Expression (10) is the same as the resistance value R_{dig} in Expression (8). The current measurement unit 63 measures the current value I_{dig} of the power source line 31 and the current value I_{ana} of the power source line 32 and outputs the current value

Idig and the current value Iana to the resistance calculation unit **64**. This current value Idig is used in Expression (10). A value Vtar_dig in Expression (10) indicates a target value (for example, 1 V) of the power source voltage Vout_dig. A resistance value Rg in Expression (10) is the same as the resistance value Rg in Expression (7). A current value Ig in Expression (10) is calculated by adding the current value Idig and the current value Iana. The power source control unit **65** adjusts the value Vout_digc of the power source voltage Vout_dig such that the value of the distal end voltage Vcis_dig matches the target value Vtar_dig. The adjustment of the value Vout_digc is repeated, and the value of the distal end voltage Vcis_dig nears the target value Vtar_dig.

(77) The following Expression (11) indicates a value Vout_anac of the power source voltage Vout_ana. The power source control unit **65** adjusts the value Vout_anac of the power source voltage Vout_ana in accordance with Expression (11) in Step **S160**.

$$V_{\text{sub.out_anac}} = R_{\text{sub.ana}} I_{\text{sub.ana}} + V_{\text{sub.tar_ana}} + R_{\text{sub.g}} I_{\text{sub.g}} \quad (11)$$

(78) A resistance value Rana in Expression (11) is the same as the resistance value Rana in Expression (9). The current value Iana measured by the current measurement unit **63** is used in Expression (11). A value Vtar_ana in Expression (11) indicates a target value (for example, 2.7 V) of the power source voltage Vout_ana. The value Vtar_ana in Expression (11) is different from the value Vtar_dig in Expression (10). A resistance value Rg in Expression (11) is the same as the resistance value Rg in Expression (7). A current value Ig in Expression (11) is calculated by adding the current value Idig and the current value Iana. The power source control unit **65** adjusts the value Vout_anac of the power source voltage Vout_ana such that the value of the distal end voltage Vcis_ana matches the target value Vtar_ana. The target value Vtar_ana is different from the target value Vtar_dig. The adjustment of the value Vout_anac is repeated, and the value of the distal end voltage Vcis_ana nears the target value Vtar_ana.

(79) The order in which Step **S105** and Step **S110** are executed may be different from that shown in FIG. **3**. For example, after Step **S110** is executed, Step **S105** may be executed. After Step **S115** is executed, at least one of Steps **S105** and **S110** may be executed.

(80) Step **S120** may be executed at any timing after Step **S110** is executed and before Step **S125** is executed.

(81) The order in which Step **S130** and Step **S135** are executed may be different from that shown in FIG. **3**. For example, after Step **S135** is executed, Step **S130** may be executed. After Step **S140** is executed, at least one of Steps **S130** and **S135** may be executed.

(82) Step **S145** may be executed at any timing after Step **S135** is executed and before Step **S150** is executed.

(83) FIG. **5** shows a procedure of an operation of the camera device **9** related to adjustment of the power source voltage Vout_dig and the power source voltage Vout_ana. The operation of the camera device **9** will be described by using FIG. **5**.

(84) The AD conversion unit **92** performs AD conversion on the distal end voltage Vcis_dig in the first period and generates a digital signal indicating a value Vcis_dig1 of the distal end voltage Vcis_dig. The AD conversion unit **92** performs AD conversion on the distal end voltage Vcis_ana in the first period and generates a digital signal indicating a value Vcis_ana1 of the distal end voltage Vcis_ana (Step **S200**).

(85) After Step **S200**, the signal-processing unit **93** outputs each digital signal to the video signal line **30** (Step **S205**).

(86) After Step **S205**, the AD conversion unit **92** performs AD conversion on the distal end voltage Vcis_dig in the second period and generates a digital signal indicating a value Vcis_dig2 of the distal end voltage Vcis_dig (Step **S210**).

(87) After Step **S210**, the signal-processing unit **93** outputs the digital signal to the video signal line **30** (Step **S215**).

(88) Each aspect of the present invention may include the following modified example. The image sensor **90** receives, as a distal end voltage Vcis_dig (first distal end voltage), a power source

voltage Vout_dig (first power source voltage) output from the control unit **6** to the power source line **31** (first power source line). The image sensor **90** receives, as a distal end voltage Vcis_ana (second distal end voltage), a power source voltage Vout_ana (second power source voltage) output from the control unit **6** to the power source line **32** (second power source line). The value of the power source voltage Vout_ana is different from that of the power source voltage Vout_dig. The AD conversion unit **92** measures a value of the distal end voltage Vcis_dig and a value of the distal end voltage Vcis_ana. The signal-processing unit **93** outputs a first digital signal indicating the value of the distal end voltage Vcis_dig and a second digital signal indicating the value of the distal end voltage Vcis_ana to the video signal line **30**. The power source unit **60** generates the power source voltage Vout_dig and the power source voltage Vout_ana. The power source unit **60** outputs the generated power source voltage Vout_dig to the power source line **31** and outputs the generated power source voltage Vout_ana to the power source line **32**. The current measurement unit **63** measures a current value of the power source line **31** and a current value of the power source line **32**. The AFE unit **50** receives the first digital signal and the second digital signal. The resistance calculation unit **64** calculates a resistance value of the power source line **31** based on the value of the power source voltage Vout_dig, the value of the distal end voltage Vcis_dig indicated by the first digital signal, the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33** (Expression (8)). The resistance calculation unit **64** calculates a resistance value of the power source line **32** based on the value of the power source voltage Vout_ana, the value of the distal end voltage Vcis_ana indicated by the second digital signal, the current value of the power source line **32**, the resistance value of the ground line **33**, and the current value of the ground line **33** (Expression (9)). The power source control unit **65** adjusts the value of the power source voltage Vout_dig based on the resistance value of the power source line **31**, the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33** such that the value of the distal end voltage Vcis_dig nears a first target value (Expression (10)). The power source control unit **65** adjusts the value of the power source voltage Vout_ana based on the resistance value of the power source line **32**, the current value of the power source line **32**, the resistance value of the ground line **33**, and the current value of the ground line **33** such that the value of the distal end voltage Vcis_ana nears a second target value (Expression (11)).

(89) Each aspect of the present invention may include the following modified example. The resistance calculation unit **64** calculates the resistance value of the ground line **33** based on the value of the power source voltage Vout_dig (first power source voltage), the value of the distal end voltage Vcis_dig (first distal end voltage) indicated by the first digital signal, and the current value of the ground line **33** (Expression (7)).

(90) Each aspect of the present invention may include the following modified example. The power source unit **60** outputs the power source voltage Vout_dig (first power source voltage) to the power source line **31** (first power source line) in a first period and a second period and outputs the power source voltage Vout_ana (second power source voltage) to the power source line **32** (second power source line) in the first period and the second period. A current value of the power source line **31** in the second period is the same as that of the power source line **31** in the first period. A current value of the power source line **32** in the second period is different from that of the power source line **32** in the first period. The resistance calculation unit **64** calculates the resistance value of the ground line **33** based on the value of the power source voltage Vout_dig in each of the first period and the second period, the value of the distal end voltage Vcis_dig (first distal end voltage) indicated by the first digital signal in each of the first period and the second period, and the current value of the ground line **33** in each of the first period and the second period (Expression (7)).

(91) Each aspect of the present invention may include the following modified example. The first period and the second period are included in a blanking period of the image sensor **90**.

(92) Each aspect of the present invention may include the following modified example. The

resistance calculation unit **64** calculates the current value of the ground line **33** based on the current value of the power source line **31** (first power source line) and the current value of the power source line **32** (second power source line).

(93) Each aspect of the present invention may include the following modified example. One of the distal end voltage V_{cis_dig} (first distal end voltage) and the distal end voltage V_{cis_ana} (second distal end voltage) is provided to a digital circuit included in the image sensor **90**. The other of the distal end voltage V_{cis_dig} and the distal end voltage V_{cis_ana} is provided to an analog circuit included in the image sensor **90**.

(94) In the first embodiment, the endoscope system **1** calculates the resistance value of the power source line **31** based on the value of the power source voltage V_{out_dig} , the value of the distal end voltage V_{cis_dig} , the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33**. In addition, the endoscope system **1** calculates the resistance value of the power source line **32** based on the value of the power source voltage V_{out_ana} , the value of the distal end voltage V_{cis_ana} , the current value of the power source line **32**, the resistance value of the ground line **33**, and the current value of the ground line **33**. Since the resistance value and the current value of the ground line **33** are considered, the endoscope system **1** can accurately calculate the resistance value of the power source line **31** and the resistance value of the power source line **32**.

(95) The endoscope system **1** adjusts the value of the power source voltage V_{out_dig} based on the resistance value of the power source line **31**, the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33**. In addition, the endoscope system **1** adjusts the value of the power source voltage V_{out_ana} based on the resistance value of the power source line **32**, the current value of the power source line **32**, the resistance value of the ground line **33**, and the current value of the ground line **33**. Since the resistance value and the current value of the ground line **33** are considered, the endoscope system **1** can accurately adjust the value of the power source voltage V_{out_dig} and the value of the power source voltage V_{out_ana} .

Modified Example of First Embodiment

(96) A modified example of the first embodiment of the present invention will be described. In the modified example of the first embodiment, a method of calculating a resistance value of the ground line **33** is different from that in the first embodiment.

(97) FIG. **6** shows a procedure of an operation of the control unit **6** related to adjustment of the power source voltage V_{out_dig} and the power source voltage V_{out_ana} . The operation of the control unit **6** will be described by using FIG. **6**. Descriptions of the same processing as that shown in FIG. **3** will be omitted.

(98) After Step **S125**, the voltage measurement unit **62** measures a value V_{out_ana2} of the power source voltage V_{out_ana} in the power source line **32** and outputs the value V_{out_ana2} to the resistance calculation unit **64** (Step **S130a**). After Step **S130a**, Step **S135** is executed.

(99) After Step **S135**, the AFE unit **50** receives a digital signal transferred by the video signal line **30** and processes the digital signal. The digital signal indicates a value V_{cis_ana2} of the distal end voltage V_{cis_ana} in the second period. The digital signal is output to the signal-processing unit **51**. The signal-processing unit **51** processes the digital signal and outputs the digital signal to the resistance calculation unit **64** (Step **S140a**). After Step **S140a**, Step **S145** is executed.

(100) FIG. **7** shows waveforms of the current I_{dig} , the current I_{ana} , and the current I_g . The horizontal axis in FIG. **7** indicates time, and the vertical axis in FIG. **7** indicates a current value.

(101) The power source unit **60** outputs the power source voltage V_{out_ana} to the power source line **32** such that a value I_{ana1} of the current I_{ana} in a first period **T1** and a value I_{ana2} of the current I_{ana} in a second period **T2** are the same. For example, a value V_{out_ana2} of the power source voltage V_{out_ana} in the second period **T2** is the same as a value V_{out_ana1} of the power source voltage V_{out_ana} in the first period **T1**.

(102) The power source unit **60** outputs the power source voltage Vout_dig to the power source line **31** such that a value Idig1 of the current Idig in the first period T1 and a value Idig2 of the current Idig in the second period T2 are different. For example, a value Vout_dig2 of the power source voltage Vout_dig in the second period T2 is different from a value Vout_dig1 of the power source voltage Vout_dig in the first period T1.

(103) In the example shown in FIG. 7, the value Idig2 of the current Idig in the second period T2 is greater than the value Idig1 of the current Idig in the first period T1. The value Idig2 of the current Idig in the second period T2 may be less than the value Idig1 of the current Idig in the first period T1.

(104) The current Ig is the sum of the current Idig and the current Iana. The value of the current Ig in the first period T1 is Ig1, and the value of the current Ig in the second period T2 is Ig2.

(105) After Step S145, the resistance calculation unit **64** calculates a resistance value Rg of the ground line **33** based on the values Vout_ana1 and Vout_ana2 of the power source voltage Vout_ana, the values Vcis_ana1 and Vcis_ana2 of the distal end voltage Vcis_ana, and the current values Ig1 and Ig2 of the ground line **33** (Step S150a). After Step S150a, Step S155 is executed.

(106) The following Expression (12) is obtained by using a similar method to that by which Expression (6) is obtained from Expression (4) and Expression (5).

$$V_{\text{sub.out_ana1}} - V_{\text{sub.out_ana2}} =$$

$$(V_{\text{sub.cis_ana1}} - V_{\text{sub.cis_ana2}}) + R_{\text{sub.g}}(I_{\text{sub.g1}} - I_{\text{sub.g2}}) \quad (12)$$

(107) By modifying Expression (12), the following Expression (13) is obtained. The resistance calculation unit **64** calculates the resistance value Rg of the ground line **33** in accordance with Expression (13) in Step S150a.

$$(108) \quad R_g = \frac{(V_{\text{out_ana1}} - V_{\text{out_ana2}}) - (V_{\text{cis_ana1}} - V_{\text{cis_ana2}})}{I_{g1} - I_{g2}} \quad (13)$$

(109) FIG. 8 shows a procedure of an operation of the camera device **9** related to adjustment of the power source voltage Vout_dig and the power source voltage Vout_ana. The operation of the camera device **9** will be described by using FIG. 8. The same processing as that shown in FIG. 5 will be omitted.

(110) After Step S205, the AD conversion unit **92** performs AD conversion on the distal end voltage Vcis_ana in the second period and generates a digital signal indicating a value Vcis_ana2 of the distal end voltage Vcis_ana (Step S210a). After Step S210a, Step S215 is executed.

(111) Each aspect of the present invention may include the following modified example. The image sensor **90** receives, as a distal end voltage Vcis_ana (first distal end voltage), a power source voltage Vout_ana (first power source voltage) output from the control unit **6** to the power source line **32** (first power source line). The image sensor **90** receives, as a distal end voltage Vcis_dig (second distal end voltage), a power source voltage Vout_dig (second power source voltage) output from the control unit **6** to the power source line **31** (second power source line). The value of the power source voltage Vout_dig is different from that of the power source voltage Vout_ana. The AD conversion unit **92** measures a value of the distal end voltage Vcis_ana and a value of the distal end voltage Vcis_dig. The signal-processing unit **93** outputs a first digital signal indicating the value of the distal end voltage Vcis_ana and a second digital signal indicating the value of the distal end voltage Vcis_dig to the video signal line **30**. The power source unit **60** generates the power source voltage Vout_ana and the power source voltage Vout_dig. The power source unit **60** outputs the generated power source voltage Vout_ana to the power source line **32** and outputs the generated power source voltage Vout_dig to the power source line **31**. The current measurement unit **63** measures a current value of the power source line **32** and a current value of the power source line **31**. The AFE unit **50** receives the first digital signal and the second digital signal. The resistance calculation unit **64** calculates a resistance value of the power source line **32** based on the value of the power source voltage Vout_ana, the value of the distal end voltage Vcis_ana indicated by the first digital signal, the current value of the power source line **32**, the resistance value of the

ground line **33**, and the current value of the ground line **33** (Expression (9)). The resistance calculation unit **64** calculates a resistance value of the power source line **31** based on the value of the power source voltage V_{out_dig} , the value of the distal end voltage V_{cis_dig} indicated by the second digital signal, the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33** (Expression (8)). The power source control unit **65** adjusts the value of the power source voltage V_{out_ana} based on the resistance value of the power source line **32**, the current value of the power source line **32**, the resistance value of the ground line **33**, and the current value of the ground line **33** such that the value of the distal end voltage V_{cis_ana} nears a first target value (Expression (11)). The power source control unit **65** adjusts the value of the power source voltage V_{out_dig} based on the resistance value of the power source line **31**, the current value of the power source line **31**, the resistance value of the ground line **33**, and the current value of the ground line **33** such that the value of the distal end voltage V_{cis_dig} nears a second target value (Expression (10)).

(112) Each aspect of the present invention may include the following modified example. The resistance calculation unit **64** calculates a resistance value of the ground line **33** based on the value of the power source voltage V_{out_ana} (first power source voltage), the value of the distal end voltage V_{cis_ana} (first distal end voltage) indicated by the first digital signal, and the current value of the ground line **33** (Expression (13)).

(113) Each aspect of the present invention may include the following modified example. The power source unit **60** outputs the power source voltage V_{out_ana} (first power source voltage) to the power source line **32** (first power source line) in a first period and a second period and outputs the power source voltage V_{out_dig} (second power source voltage) to the power source line **31** (second power source line) in the first period and the second period. A current value of the power source line **32** in the second period is the same as that of the power source line **32** in the first period. A current value of the power source line **31** in the second period is different from that of the power source line **31** in the first period. The resistance calculation unit **64** calculates the resistance value of the ground line **33** based on the value of the power source voltage V_{out_ana} in each of the first period and the second period, the value of the distal end voltage V_{cis_ana} (first distal end voltage) indicated by the first digital signal in each of the first period and the second period, and the current value of the ground line **33** in each of the first period and the second period (Expression (13)).

(114) Each aspect of the present invention may include the following modified example. The first period and the second period are included in a blanking period of the image sensor **90**.

(115) Each aspect of the present invention may include the following modified example. The resistance calculation unit **64** calculates the current value of the ground line **33** based on the current value of the power source line **32** (first power source line) and the current value of the power source line **31** (second power source line).

(116) In the modified example of the first embodiment, the endoscope system **1** can accurately calculate the resistance value of the power source line **31** and the resistance value of the power source line **32** as in the first embodiment. In addition, the endoscope system **1** can accurately adjust the value of the power source voltage V_{out_dig} and the value of the power source voltage V_{out_ana} as in the first embodiment.

Second Embodiment

(117) A second embodiment of the present invention will be described. In the second embodiment, the endoscope system **1** shown in FIG. 1 and FIG. 2 is used.

(118) The conductivity of the ground line **33** is the same as that of the power source line **31**. The diameter of the ground line **33** is the same as that of the power source line **31**. The length of the ground line **33** is the same as that of the power source line **31**. If these conditions are met, the resistance value of the ground line **33** may be regarded as being the same as that of the power source line **31**.

(119) FIG. 9 shows a procedure of an operation of the control unit **6** related to adjustment of the

power source voltage Vout_dig and the power source voltage Vout_ana. The operation of the control unit **6** under the condition that the resistance value of the ground line **33** is the same as that of the power source line **31** will be described by using FIG. **9**. Descriptions of the same processing as that shown in FIG. **3** will be omitted.

(120) After Step **S120**, the resistance calculation unit **64** calculates the resistance value Rdig of the power source line **31** based on the value of the power source voltage Vout_dig, the value of the distal end voltage Vcis_dig, the current value of the power source line **31**, and the current value Ig of the ground line **33**. In addition, the resistance calculation unit **64** calculates the resistance value Rana of the power source line **32** based on the value of the power source voltage Vout_ana, the value of the distal end voltage Vcis_ana, the current value of the power source line **32**, and the current value Ig of the ground line **33**. The resistance calculation unit **64** outputs the resistance value Rdig of the power source line **31** and the resistance value Rana of the power source line **32** to the power source control unit **65** (Step **S155b**). After Step **S155b**, Step **S160** is executed.

(121) When the resistance value of the ground line **33** is the same as that of the power source line **31**, Expression (8) described above is changed to the following Expression (14). The resistance calculation unit **64** calculates the resistance value Rdig of the power source line **31** in accordance with Expression (14) in Step **S155b**.

$$(122) \quad R_{\text{dig}} = \frac{V_{\text{out_dig1}} - V_{\text{cis_dig1}}}{I_{\text{dig1}} + I_{g1}} \quad (14)$$

(123) The resistance calculation unit **64** calculates the resistance value Rana of the power source line **32** in accordance with Expression (9) described above in Step **S155b**. The resistance value Rg in Expression (9) is the same as the resistance value Rdig of the power source line **31**.

(124) The camera device **9** executes Step **S200** and Step **S205** shown in FIG. **5**, but does not need to execute Step **S210** and Step **S215** shown in FIG. **5**.

(125) The resistance calculation unit **64** may calculate a ratio Kdig (=Rg/Rdig) between the resistance value Rg of the ground line **33** and the resistance value Rdig of the power source line **31** based on the conductivity of the ground line **33**, the conductivity of the power source line **31**, the diameter of the ground line **33**, the diameter of the power source line **31**, the length of the ground line **33**, and the length of the power source line **31**. The resistance calculation unit **64** may calculate the resistance value Rdig of the power source line **31** in accordance with the following Expression (15) in Step **S155b**.

$$(126) \quad R_{\text{dig}} = \frac{V_{\text{out_dig1}} - V_{\text{cis_dig1}}}{I_{\text{dig1}} + K_{\text{dig}} I_{g1}} \quad (15)$$

(127) In the second embodiment, the endoscope system **1** does not need to execute the processing in the second period described above. Therefore, the endoscope system **1** can shorten the processing time required for adjusting the power source voltage Vout_dig and the power source voltage Vout_ana.

Modified Example of Second Embodiment

(128) A modified example of the second embodiment of the present invention will be described. The conductivity of the ground line **33** is the same as that of the power source line **32**. The diameter of the ground line **33** is the same as that of the power source line **32**. The length of the ground line **33** is the same as that of the power source line **32**. If these conditions are met, the resistance value of the ground line **33** may be regarded as being the same as that of the power source line **32**.

(129) When the resistance value of the ground line **33** is the same as that of the power source line **32**, Expression (9) described above is changed to the following Expression (16). The resistance calculation unit **64** calculates the resistance value Rana of the power source line **32** in accordance with Expression (16) in Step **S155b**.

$$(130) \quad R_{\text{ana}} = \frac{V_{\text{out_ana1}} - V_{\text{cis_ana1}}}{I_{\text{ana1}} + I_{g1}} \quad (16)$$

(131) The resistance calculation unit **64** calculates the resistance value Rdig of the power source line **31** in accordance with Expression (8) described above in Step **S155b**. The resistance value Rg

in Expression (8) is the same as the resistance value Rana of the power source line 32.

(132) The resistance calculation unit 64 may calculate a ratio Kana (=Rg/Rana) between the resistance value Rg of the ground line 33 and the resistance value Rana of the power source line 32 based on the conductivity of the ground line 33, the conductivity of the power source line 32, the diameter of the ground line 33, the diameter of the power source line 32, the length of the ground line 33, and the length of the power source line 32. The resistance calculation unit 64 may calculate the resistance value Rana of the power source line 32 in accordance with the following Expression (17) in Step S155b.

$$(133) \quad R_{ana} = \frac{V_{out_ana1} - V_{cis_ana1}}{I_{ana1} + K_{ana} I_{g1}} \quad (17)$$

(134) In the modified example of the second embodiment, the endoscope system 1 can shorten the processing time required for adjusting the power source voltage Vout_dig and the power source voltage Vout_ana as in the second embodiment described above.

(135) While preferred embodiments of the invention have been described and shown above, it should be understood that these are examples of the invention and are not to be considered as limiting. Additions, omissions, substitutions, and other modifications can be made without departing from the spirit or scope of the present invention. Accordingly, the invention is not to be considered as being limited by the foregoing description, and is only limited by the scope of the appended claims.

Claims

1. An imaging system, comprising a camera device and a control device connected to each other by a power source line, a video signal line, and a ground line, the camera device including: an image sensor that receives, as a distal end voltage, a power source voltage output from the control device to the power source line and generates a video signal by using the distal end voltage; a voltage measurement circuit that measures a value of the distal end voltage; and an output circuit that outputs the video signal to the video signal line and outputs a digital signal indicating the value of the distal end voltage to the video signal line, and the control device including: a voltage generation circuit that generates the power source voltage and outputs the generated power source voltage to the power source line; a current measurement circuit that measures a current value of the power source line; a reception circuit that receives the video signal and the digital signal output from the camera device to the video signal line; a calculation circuit that calculates a resistance value of the power source line based on a value of the power source voltage, the value of the distal end voltage indicated by the digital signal, the current value of the power source line, a resistance value of the ground line, and a current value of the ground line; and a voltage adjustment circuit that adjusts a value of the power source voltage based on the resistance value of the power source line, the current value of the power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the distal end voltage nears a target value.
2. The imaging system according to claim 1, wherein the power source line includes a first power source line and a second power source line, wherein the image sensor receives, as a first distal end voltage, a first power source voltage output from the control device to the first power source line and receives, as a second distal end voltage, a second power source voltage output from the control device to the second power source line, wherein a value of the second power source voltage is different from a value of the first power source voltage, wherein the voltage measurement circuit measures a value of the first distal end voltage and a value of the second distal end voltage, wherein the output circuit outputs a first digital signal indicating the value of the first distal end voltage and a second digital signal indicating the value of the second distal end voltage to the video signal line, wherein the voltage generation circuit generates the first power source voltage and the second power source voltage, outputs the generated first power source voltage to the first power

source line, and outputs the generated second power source voltage to the second power source line, wherein the current measurement circuit measures a current value of the first power source line and a current value of the second power source line, wherein the reception circuit receives the first digital signal and the second digital signal, wherein the calculation circuit calculates a resistance value of the first power source line based on the value of the first power source voltage, the value of the first distal end voltage indicated by the first digital signal, the current value of the first power source line, the resistance value of the ground line, and the current value of the ground line, wherein the calculation circuit calculates a resistance value of the second power source line based on the value of the second power source voltage, the value of the second distal end voltage indicated by the second digital signal, the current value of the second power source line, the resistance value of the ground line, and the current value of the ground line, wherein the voltage adjustment circuit adjusts the value of the first power source voltage based on the resistance value of the first power source line, the current value of the first power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the first distal end voltage nears a first target value, and wherein the voltage adjustment circuit adjusts the value of the second power source voltage based on the resistance value of the second power source line, the current value of the second power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the second distal end voltage nears a second target value.

3. The imaging system according to claim 2, wherein the calculation circuit calculates the resistance value of the ground line based on the value of the first power source voltage, the value of the first distal end voltage indicated by the first digital signal, and the current value of the ground line.

4. The imaging system according to claim 2, wherein the voltage generation circuit outputs the first power source voltage to the first power source line in a first period and a second period and outputs the second power source voltage to the second power source line in the first period and the second period, wherein a current value of the first power source line in the second period is the same as a current value of the first power source line in the first period, wherein a current value of the second power source line in the second period is different from a current value of the second power source line in the first period, and wherein the calculation circuit calculates the resistance value of the ground line based on the value of the first power source voltage in each of the first period and the second period, the value of the first distal end voltage indicated by the first digital signal in each of the first period and the second period, and the current value of the ground line in each of the first period and the second period.

5. The imaging system according to claim 4, wherein the first period and the second are included in a blanking period of the image sensor.

6. The imaging system according to claim 2, wherein the calculation circuit calculates the current value of the ground line based on the current value of the first power source line and the current value of the second power source line.

7. The imaging system according to claim 2, wherein one of the first distal end voltage and the second distal end voltage is provided to a digital circuit included in the image sensor, and wherein the other of the first distal end voltage and the second distal end voltage is provided to an analog circuit included in the image sensor.

8. The imaging system according to claim 1, wherein the resistance value of the ground line is the same as the resistance value of the power source line.

9. An endoscope system, comprising a scope to be inserted into a living body; and the imaging system according to claim 1, wherein the camera device is disposed in a distal end of the scope.

10. A control device in an imaging system including a camera device and the control device connected to each other by a power source line, a video signal line, and a ground line, the control device comprising: a voltage generation circuit that generates a power source voltage and outputs

the generated power source voltage to the power source line; a current measurement circuit that measures a current value of the power source line; a reception circuit that receives a video signal and a digital signal output from the camera device to the video signal line, wherein the video signal is generated by an image sensor of the camera device, wherein the image sensor receives, as a distal end voltage, the power source voltage output from the control device to the power source line and generates the video signal by using the distal end voltage, and wherein the digital signal indicates a value of the distal end voltage; a calculation circuit that calculates a resistance value of the power source line based on a value of the power source voltage, the value of the distal end voltage indicated by the digital signal, the current value of the power source line, a resistance value of the ground line, and a current value of the ground line; and a voltage adjustment circuit that adjusts a value of the power source voltage based on the resistance value of the power source line, the current value of the power source line, the resistance value of the ground line, and the current value of the ground line such that the value of the distal end voltage nears a target value.
